# 12.5 Functions of three variables



We now go from two input variables to three: $w = f(x,y,z)$. Unless you can see into four dimensions (you can't) you cannot visualize the graph of $f$.

Cross sections of $f$ with $w$ fixed are called level surfaces.

→ Do problems 1-6.

So, surfaces are used to visualize $z = f(x,y)$ as a graph. They are also used to visualize $w = f(x,y,z)$ as level surfaces.

**Fact.** Every graph of a function $f(x,y)$ appears as a level surface of a function $g(x,y,z)$. Just take:

$$g(x,y,z) = f(x,y) - z.$$

Then $f$ appears as the level surface $g(x,y,z) = c$ with $c = 0$. Sometimes it's possible to go the other way: from level surfaces to graphs.

→ Do problems 7 and 8.